



Project Report: Café Sales & Operational Analytics Dashboard

1. Project Overview & Problem Statement

- **Background:** Independent food and beverage retail businesses often struggle to track holistic operational performance due to fragmented data across multiple transactions, varying payment channels, and diverse dining preferences.
- **The Problem:** Management lacked a centralized, data-driven solution to monitor overall business performance, understand customer purchasing patterns, and evaluate item-level profitability across **10,000 retail transactions generating \$89.19K in total sales**. Furthermore, high volumes of unrecorded operational metrics ("Unspecified" POS logs) presented a challenge in tracking true customer behavior during peak hours.
- **Objective:** To design and develop an end-to-end analytics solution—from raw data processing in Python to an interactive Power BI dashboard—providing actionable insights into temporal sales trends, category preferences, pricing structures, and operational bottlenecks.

2. End-to-End Analytical Process

Phase 1: Data Preparation & Cleaning (Python)

Before building the dashboard, the raw dataset underwent rigorous data cleaning and preprocessing using Python:

- **Handling Missing Values:** Addressed nulls and inconsistencies in text-based categorical columns, standardizing naming conventions for items, categories, and payment methods.
- **Data Type Formatting:** Converted date fields into proper datetime objects to enable smooth temporal aggregation and chronological sorting.
- **Feature Engineering:** Ensured numeric fields like *Price Per Unit*, *Quantity*, and *Total Sales* were correctly typed to support precise metric calculations.

Phase 2: Data Modeling & Visualization (Power BI)

- **Design Aesthetic:** Developed a custom "Warm Café" theme utilizing rich espresso browns (#2C1D11), creams, and warm caramels to establish a professional, brand-aligned visual identity.
- **Data Modeling & Sorting:** Established proper relationships and utilized the "Sort by Column" feature to ensure months sorted chronologically (January to December) rather than alphabetically.
- **Visual Hierarchy:** Built a clean layout incorporating high-level KPI cards (Total Sales, Total Quantity, Average Price), temporal line charts, category bar charts, and operational donut charts.

3. Key Insights & Analytical Findings

- **Payment Preference:** Customers heavily favor digital/online payment methods over cash or offline alternatives, highlighting a strong consumer shift toward cashless transactions.
- **Category Dominance:** "Eatables" (such as salads, cakes, and sandwiches) are the highest-selling product category, outperforming beverage sales overall.
- **Top Revenue-Generating Item:** **Salads** stand out as the single most sold and highest-revenue-generating item on the menu.
- **Temporal Dip in Sales:** **February** records the lowest sales performance of the year, providing a clear window for targeted promotions or seasonal warm-up specials during late winter.
- **Dining Options Split:** Customer preference is remarkably balanced between dining in-store and opting for takeaway, showing that the café serves both as a sit-down spot and a quick-service stop.
- **Weekend vs. Weekday Traffic:** Weekend volume spikes visibly—with **Saturdays and Sundays** driving heavier transaction values (crossing over \$500 blocks), while weekdays maintain a steady, moderate baseline of sales.

4. Operational Impact & Data Quality Analysis

- **The "Unspecified" Data Dilemma:** A significant portion of transactions lacked specific logging for payment categories and dining options, leaving large gaps recorded as "Unspecified."

- **Root Cause Diagnosis:** Rather than treating this as dirty data, it was framed as an operational insight: fast-paced, high-volume rush hours cause staff to intentionally skip optional fields on the Point-of-Sale (POS) terminal just to keep lines moving and serve customers faster.
- **Business Consequence:** While checkout speed is preserved, management loses clear visibility into true customer segmentation and exact channel preferences.

5. Strategic Recommendations & Action Plan

- **Fix the Checkout Data:** Update the point-of-sale (POS) system with fast, easy buttons so cashiers can quickly log how customers pay and where they eat without slowing down the line.
- **Boost Slow Months:** Run special meal bundles or loyalty discounts in **February** to bring in more customers during the slowest month of the year.
- **Prep for Weekends:** Schedule extra staff and make sure popular food items like salads and sandwiches are fully stocked ahead of time to handle the heavy customer rush on **Saturdays and Sundays**.